Production I: Firms and the Shock

Advanced Microeconomics 1: Economic Decision Making. Lecture 5

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Where we left the question

- L1–L4 answered the block's questions for households: who the 2022 shock hurt, by how much, and whether one-basket indexation is the right compensation.
- But we took the price change as given, $p^0 \to p^1$ just happened. And we treated indexation purely as a payment to households.
- Both are half the story. The same shock hit firms, as an input price, and indexation raises wages, which are (part of) a firm's costs.
- This block: the supply side.

The same shock, from the other side

- Energy is an input.
- BASF — Europe's largest chemicals maker, whose Ludwigshafen site is one of the biggest gas consumers on the continent — took an €800m hit from rising gas prices in a single quarter. Its CEO: "energy prices will stay at a high level and won't go back to normal soon." Aluminium producers warned of "production cuts, job losses, and possibly a complete breakdown."
- So: did European industry collapse in 2022?

It did not

- In 2022, EU industry cut natural-gas use by 15% against the 2019–21 average.
- Over the same period, industrial output rose by around 5% (Bruegel, 2023).
- Firms cut gas, not production. They substituted away from the input that got expensive.
- Exactly analogous to a household substituting away from the expensive good.

Today's route

1. Formalise the firm.
2. The profit maximization problem: $\pi(p)$ and $y(p)$, Hotelling's lemma.
3. The law of supply.

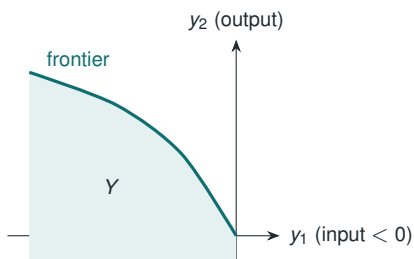
Formalising the firm

Production vector

Definition · production vector and production set

A **production vector** is $y \in \mathbb{R}^L$: $y_\ell > 0$ means output of good ℓ, $y_\ell < 0$ means input. The **production set** $Y \subseteq \mathbb{R}^L$ collects all technologically feasible y.

- A bakery:
  $y = (-\text{gas}, -\text{labour}, -\text{flour}, +\text{bread})$.
  The 2022 shock is a price change on a negative component.
- Profit is $p \cdot y$.



Properties a production set can have (MWG 5.B)

Property	Content
Y nonempty, closed	limits of feasible plans are feasible
No free lunch	no outputs without inputs
Possibility of inaction	$0 \in Y$: can shut down
Free disposal	can absorb extra inputs harmlessly
Irreversibility	can't run production backwards
Returns to scale	nonincreasing / nondecreasing / constant
Additivity	$y + y' \in Y$: replication across plants
Convexity	mixtures of feasible plans are feasible

Describing the frontier

- **Single output**: $q = f(z)$, inputs $z \in \mathbb{R}_+^{L-1}$; $Y = \{(-z, q) : q \leq f(z)\}$.

- Slope of an isoquant — the **marginal rate of technical substitution**:

$$\text{MRTS}_{\ell k}(z) = \frac{\partial f(z)/\partial z_\ell}{\partial f(z)/\partial z_k}$$

  how much more of input k you need when you give up a little of input ℓ, holding output fixed. The producer's MRS.

- Returns to scale for f: $f(\alpha z) \gtreqless \alpha f(z)$.

The profit maximization problem

How does a firm choose its production plan

- Same as in L2: hypothesise the firm picks its most profitable feasible plan.
- For the consumer this was hard: utility is invisible, choices only reveal a ranking. For the firm it is “easy”: the objective is **observable and cardinal**: Profit.
- And there is **no budget constraint**. No wealth effects, no Giffen goods, no compensation subtleties.

The profit maximization problem

PMP

Given $p \gg 0$ and Y (nonempty, closed, free disposal):

$$\pi(p) = \max_{y \in Y} \; p \cdot y, \qquad y(p) = \{y \in Y : p \cdot y = \pi(p)\}.$$

- π: **profit function**; $y(p)$: **supply correspondence**.

Properties of π and y

Proposition · MWG 5.C.1

For Y closed with free disposal:

1. π is **homogeneous of degree 1**; y is **h.d.0**.
2. π is **convex** in p.
3. **Hotelling's lemma**: if $y(\bar{p})$ is a singleton, $\nabla\pi(\bar{p}) = y(\bar{p})$.
4. If y is differentiable: $Dy(\bar{p}) = D^2\pi(\bar{p})$ is **symmetric, positive semidefinite**, $Dy(\bar{p})\,\bar{p} = 0$.

Proofs: MWG 5.C — each is the mirror of one from consumer theory.

The law of supply

Proposition · law of supply

For any p, p' and $y \in y(p)$, $y' \in y(p')$:

$$(p - p') \cdot (y - y') \geq 0.$$

Proof. Optimality at p: $p \cdot y \geq p \cdot y'$. Optimality at p': $p' \cdot y' \geq p' \cdot y$. Add and rearrange. □

- Output rises when its price rises; input demand falls when its price rises. Gas gets more expensive ⇒ use less gas.
- And no compensation required — contrast to the law of demand which held only for compensated changes. Why: no budget constraint ⇒ no income effect to fight the substitution effect.

Testing profit maximisation I: smooth data

Suppose we observe a differentiable supply function $y(p)$ at *every* $p \gg 0$.

Is there a production set Y (closed, free disposal) such that $y(p)$ solves the PMP at every p?

Proposition · firm-side integrability

$y(\cdot)$ is rationalisable by some $Y \iff$ for all p:

1. y is **h.d.0** in p;
2. $Dy(p)$ is **symmetric**;
3. $Dy(p)$ is **positive semidefinite**.

The catch: we need the response at every price vector. Real data are a handful of observations.

Compare L4: Slutsky symmetry + NSD + h.d.0 $\iff$ demand comes from some rational preference.

Testing profit maximisation II: finite data

Data: T observations (p^t, y^t), $t = 1, \ldots, T$.

Y **rationalises** the data if, for every t, $y^t \in Y$ and $p^t \cdot y^t \geq p^t \cdot y$ for all $y \in Y$.

Definition · WAPM (Varian 1984)

$$p^t \cdot y^t \;\geq\; p^t \cdot y^s \qquad \text{for all } s, t.$$

Every observed plan beats every other observed plan *at its own prices*.

Proposition · WAPM is necessary

If some Y rationalises the data, the data satisfy WAPM.

Proof. $y^s \in Y$, and y^t is optimal in Y at p^t, so $p^t \cdot y^t \geq p^t \cdot y^s$. $\square$

Finite data: WAPM is also sufficient

Theorem · Varian (1984)

The data are rationalisable $\iff$ WAPM holds $\iff$ the data are rationalisable by a **closed, convex, free-disposal** Y.

Back to the headlines

- **Gas.** Only the gas price moved a lot, so the law of supply says gas use falls. A 15% cut is what an optimising firm does, not a sign of collapse.
- **Output.** The theory does *not* deliver "output up". Output responds to its own price and to marginal cost; with marginal cost up, output should fall, unless demand shifted.
- So the PMP explains half of the 2022 pattern.

To separate *economising on gas* from *scaling back output*, and to say what indexed wages do to costs, prices and jobs, we need the cost function.

Where we stand

Takeaways

- **Testable without compensation:** symmetric PSD Dy with smooth data; WAPM with finite data.
- **The gas cut** is the law of supply at work. **The output rise** is not explained yet.
- **Ahead:** Thursday, $c(w, q)$ — the EMP's twin.